# Instructor Guide: Build the OrbitCo Operational Dashboard

**Type:** Hands-on lab (individual work) **Time:** 25 minutes (20 min individual work + 5 min share-out)

## Pre-Activity Setup

- Confirm every learner has a completed Lesson 1 workbook (`orbitco-workbook-[yourname].xlsx`) with the `Tickets`, `Uptime`, `SLALookup`, and `outputs` sheets in place. Any learner who did not finish Lesson 1 should receive the instructor-provided `orbitco-workbook-sample.xlsx` before the lab starts — this lab cannot begin from scratch.
- Project a completed reference dashboard on the screen from the first minute of the lab. The three-band layout is much easier to copy than to describe, and learners routinely try to improvise a four-band or two-band variant if they have not seen the target.
- Open your own instance of Excel on the projector so you can demonstrate the icon-set-thresholds-as-decimals step live the first time a learner asks.
- Have the `orbitco-dashboard-sample.xlsx` solution file open on your own machine for reference but not visible to learners. Use it to settle debates quickly ("that is what your chart should look like, yes") without handing it out as a template.

## Facilitation Steps

1. **0:00 — Launch** (2 min): Display the completed reference dashboard on the projector for ten seconds in silence, then hide it. Read the three-band rule aloud: "Tiles on top, trends in the middle, details on the bottom. Everything else is on another sheet." Start the timer.

2. **0:02 — Circulation, first 6 minutes** (focus: steps 1-2, tile layout): Most learners will build five tiles in a single row quickly. Watch for the two common errors — hardcoded values ("413" typed literally instead of a formula) and tiles pulling from the wrong sheet. Correct both on the spot with a one-line fix.

3. **0:08 — Circulation, middle 7 minutes** (focus: steps 3-5, pivots and charts): The pivot-table step is where learners slow down. If three or more are stuck on the "where does the PivotTable go?" question, pause the whole class for a 30-second shared demo of the **Insert → PivotTable → New Worksheet** flow. After that, they should move quickly to the line and donut charts.
4. **0:15 — Circulation, last 5 minutes** (focus: steps 6-10, polish and `README`): The traffic-light-thresholds-as-decimals step is the single most commonly-broken thing in the lab. Check every Uptime % tile personally — most will have `99.9` instead of `0.999` as the threshold and the icons will be all red or all green. The `README` tab is the step learners skip under time pressure; do not let them. The `README` is the OJL 5d artifact and the lab is incomplete without it.
5. **0:20 — Stop hands** (share-out, 5 min): Pick two learners whose dashboards display best and have them share screens for 90 seconds each. Use the debrief prompts below.

## What to Circulate For

- **Hardcoded tile values.** Learners under time pressure type "413" into the Total Tickets tile. Ask: "What happens when the `Tickets` Table grows?" If they answer "I'll update the cell," push them to a structured-reference formula instead. The dashboard has to keep working when nobody is updating it by hand.
- **Pie charts with more than five slices.** If a learner pie-charts `ResolutionCode` with eight categories instead of doing a three-slice severity donut, redirect them. Horizontal bar chart sorted by count is the honest fix for any categorical breakdown with more than five items.
- **Traffic-light thresholds in the wrong units.** The Uptime % cell contains a decimal like `0.9992`. If the thresholds are set to `99.9` the condition `cell ≥ 99.9` is false on every row and everything reads red. The fix is to enter thresholds as decimals (`0.999`, `0.995`). If the cell shows `99.92%` because it is formatted as percentage, the underlying value is still `0.9992` — format is display, not content.
- **Raw data on the dashboard sheet.** Some learners copy the `Tickets` Table onto the dashboard for the details band and the table lands above the tiles. Move it below the charts or to a separate sheet with a named-range reference. The tile strip must be at the top, and nothing above it.

- **No `README` tab.** The shortcut skip. Stop them at step 9 and make them write the five sections. If they argue about time, point out that the lab specifically lists the `README` as a deliverable and that its absence is the most common reason dashboards go stale in production.

## Common Pitfalls

- **PivotTable does not refresh.** A learner adds a test row to the `Tickets` Table, looks at the line chart, and reports that nothing changed. The pivot is stale because PivotTables do not auto-refresh. Walk them through **PivotTable Analyze → Refresh** and pin Refresh All to their Quick Access Toolbar.
- **Sparkline shows a single flat line.** The sparkline source range is pointing at a single cell or an empty range. Have them rebuild the daily-MTTR helper column on `tickets_clean` first (one row per date), then recreate the sparkline with that column as the source.
- **Conditional formatting applied to a fixed range instead of the Table column.** The red-fill-on-BREACH rule works today but fails when the Table grows because the rule stops at `$A$413`. Delete and re-apply, selecting the Table column (via the column header) before creating the rule.
- **Dashboard looks fine in Excel but breaks in PowerPoint.** Learners try to paste the whole dashboard into the Lesson 4 deck and complain that the rendering is wrong. That is a Lesson 4 problem — this lab is Excel-native. Redirect them to focus on the dashboard itself; export comes later.
- **Gridlines accidentally left on.** Easy to fix (**View → uncheck Gridlines** on the `dashboard` sheet only), but every year at least one learner forgets. Check on final submission.

## Debrief Prompts

- **"If someone looked at your dashboard for five seconds, what is the one thing they would walk away knowing?"** The answer should be a specific operational fact ("uptime is in the yellow band this month" or "high-severity tickets are concentrated in week 3"). If the answer is generic ("we had incidents"), the dashboard is doing decoration rather than communication and the tile strip needs a rework.

- **"Add a week of fake tickets to the `Tickets` Table and refresh the pivots. Did every chart update? Did every KPI tile update?"** Any charts or tiles that did not are a maintenance bug waiting to happen. Structured references on tiles and pivot-driven charts both update; hardcoded references and static chart ranges do not. This is the live demonstration of why the hygiene rules matter.
- **"If this dashboard is going to live in front of the team for a year, what needs to be on the `README` tab?"** Draw out: data source (filename, system, refresh cadence), refresh steps (what buttons to click in what order), owner (who to ask when something breaks), KPI definitions (what "MTTR" actually means in this workbook), and the change log (dates and reasons for structural edits). This is the OJL 5d conversation in miniature.

## Expected Answers

The expected end-state of the `orbitco-dashboard-[yourname].xlsx` workbook:

| Element | Expected state |
|---|---|
| `dashboard` sheet | Five tiles top, two charts middle, details band bottom, `Data as of:` top-right, gridlines off |
| `pivots` sheet | Two PivotTables (weekly count, severity-mix %) driving the two dashboard charts |
| `README` sheet | Five sections: purpose, data source, refresh steps, KPI definitions, owner |
| Uptime % traffic light | Green on values ≥ 0.999, yellow on 0.995–0.999, red otherwise |
| Details band | Red fill on SLA `BREACH` rows, bold on `High` severity rows, rules applied to the Table column |
| KPI formulas | Structured references (e.g., `=COUNTA(Tickets[TicketID])`), no hardcoded numbers |
| MTTR sparkline | Line sparkline populated from a daily-MTTR helper column |

If a learner's dashboard misses any of these, step through the Instructions list in order — omissions almost always trace to the `README` or the traffic-light thresholds (the two steps that get skipped under time pressure), not to the core layout work.